Advanced Functions & Equations

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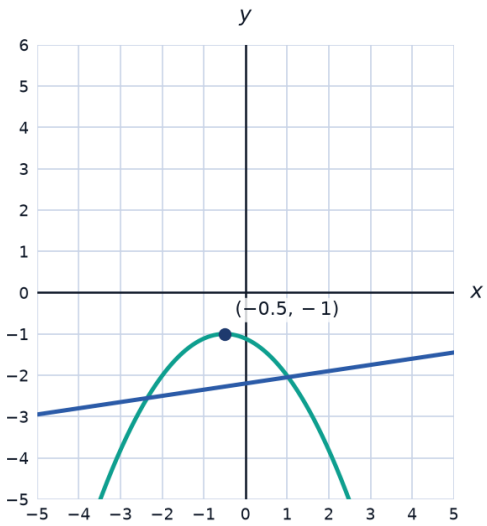
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Function composition

Using the same f and g, find the formula for $(f \circ g)(x)$.

2 [Foundational] • Target: Inverse functions

Verify your answer above by computing $f(f^{-1}(5))$.



3 [Foundational] • Target: Factoring higher-degree polynomials

Given that $x = 1$ is a root of $x^3 - 2x^2 - 5x + 6 = 0$, use the Factor Theorem to factor completely and find all roots.

4 [Foundational] • Target: Factoring higher-degree polynomials

Verify that $x = -2$ is a root of $x^3 - 2x^2 - 5x + 6 = 0$ by direct substitution.

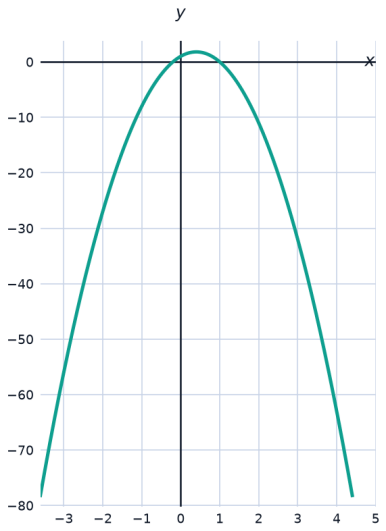
5 [Foundational] • Target: Exponential and logarithmic equations

Solve $\log_3(x + 2) + \log_3(x - 2) = 2$, checking for extraneous roots.

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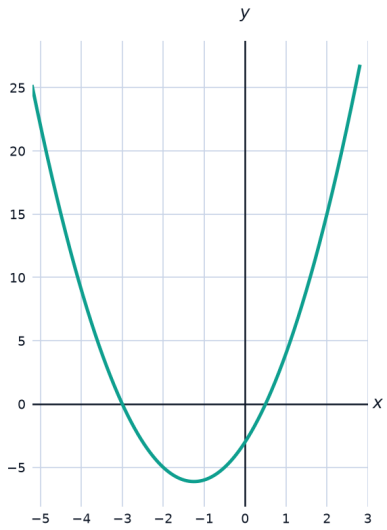
6 [Foundational] • Target: Function composition

Find the slant (oblique) asymptote of the rational function: $g(x) = \frac{2x^3 - 5x^2 + 4x + 1}{x^2 - x + 1}$



7 [Foundational] • Target: Inverse functions

Find the vertical and horizontal asymptotes of the rational function: $f(x) = \frac{4x^2 - 1}{2x^2 + 5x - 3}$



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Rational equations

Evaluate the limit of the rational expression by factoring: $\lim_{x \to 3} \frac{x^3 - 27}{x^2 - 2x - 3}$

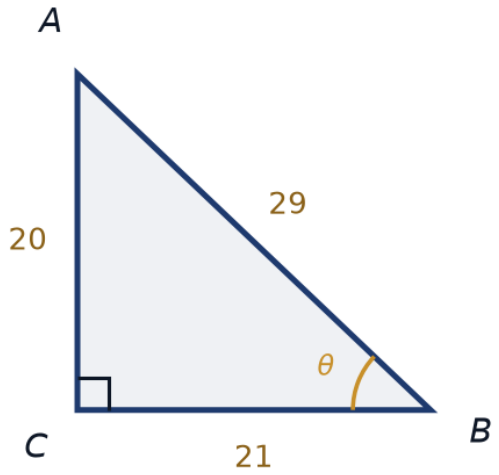
9 [Intermediate] • Target: Factoring higher-degree polynomials

Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{2x + 3}{x - 5}$ where $x \neq 5$.

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10 [Intermediate] • Target: Exponential and logarithmic equations

In triangle ABC, $a = 10$, $b = 16$, and $A = 30^\circ$. Determine whether $0, 1,$ or 2 distinct triangles can be formed, and calculate all possible values for angle B to the nearest tenth of a degree.



11 [Intermediate] • Target: Function composition

Determine the value of the constant k for which the function $f(x)$ is continuous at $x = 2$:

$f(x) = \{kx^2 + 2x, , x \leq 2; 3x + k, , x > 2$

12 [Intermediate] • Target: Inverse functions

Solve the rational equation: $\frac{2}{x-3} + \frac{1}{x} = \frac{x-1}{x-3}$

13 [Intermediate] • Target: Rational equations

Find the value of constant k such that the function $f(x)$ is continuous at $x = 3$:

$f(x) = \left\{ \frac{x^2-9}{x-3}, , xeq3; k, , x = 3 \right.$

14 [Intermediate] • Target: Factoring higher-degree polynomials

The graph of $y = f(x)$ is transformed to $y = -2f(x - 3) + 5$. Describe the sequence of geometric transformations applied to the parent function.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

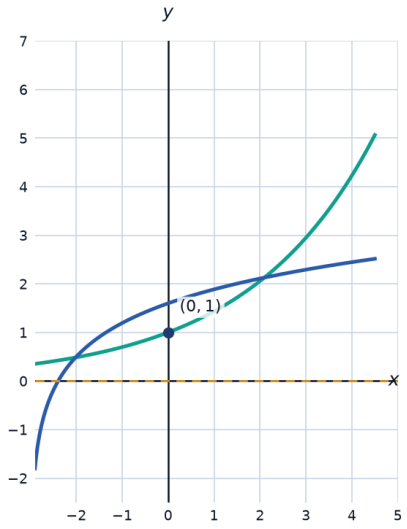
15 [Challenging] • Target: Exponential and logarithmic equations • Cross-Lesson: 3.01 Sequences & Series

If $f(x) = 2x^2 - 3$ and $g(x) = \sqrt{x+1}$, find the domain of the composite function $(g \circ f)(x)$.

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16 [Challenging] • Target: Function composition • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Solve the exponential equation for x: $4^{x+1} = 8^{2x-1}$



17 [Challenging] • Target: Inverse functions • Cross-Lesson: 6.01 Limits

[Test format & strategy · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

18 [Challenging] • Target: Rational equations • Cross-Lesson: 1.01 Complex Numbers

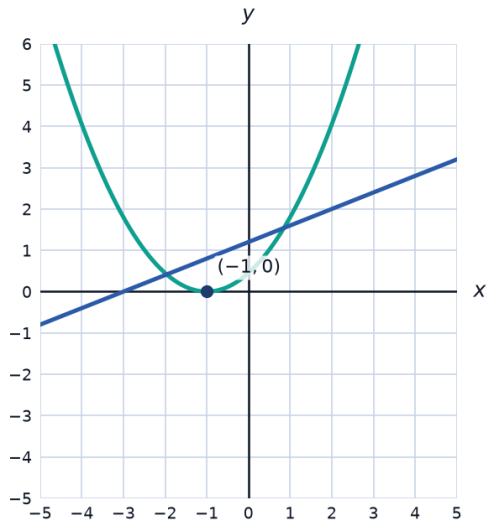
Find all vertical and horizontal asymptotes, the exact coordinates of any holes, and all intercepts of the rational function $R(x) = \frac{2x^2 + 5x - 3}{x^2 + 4x + 3}$.

19 [Challenging] • Target: Factoring higher-degree polynomials • Cross-Lesson: 2.01 Matrices & Vectors

Given $f(x) = \frac{2x + 5}{x - 3}$ for $x \neq 3$, find the inverse function $f^{-1}(x)$ and state its domain.

20 [Challenging] • Target: Exponential and logarithmic equations • Cross-Lesson: 3.01 Sequences & Series

Find the distance in the polar coordinate plane between the two points $P_1\left(4, \frac{\pi}{3}\right)$ and $P_2\left(7, \frac{2\pi}{3}\right)$.



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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Function composition

Given that $x = 4$ is a root of $x^3 - 13x - 12 = 0$, use the Factor Theorem to find the other two roots.

a. $(-3, -1)$

b. $(3, 1)$

c. $(-2, -1)$

d. $(-3, -2)$

22 [Foundational] • Target: Inverse functions

A line L is tangent to the graph of the polynomial function $f(x) = x^4 - 6x^2 + 5x + 7$ at two distinct points in the xy-plane. What is the y-intercept of line L?

a. -2

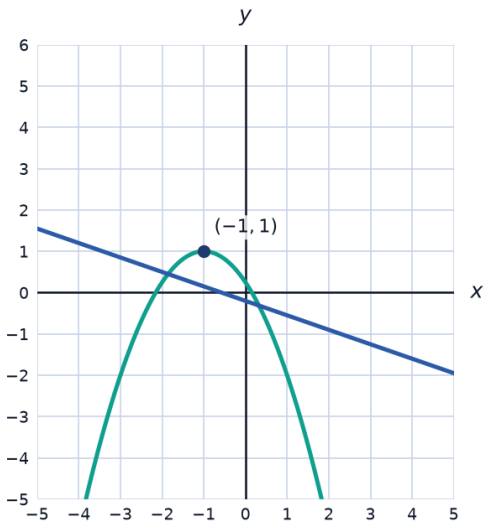
b. 2

c. 7

d. 16

23 [Foundational] • Target: Rational equations

The polynomial function $P(x) = x^3 + ax^2 + bx + 12$, where a and b are real constants, has $x - 1$ as a factor. When $P(x)$ is divided by $x - 2$, the remainder is -10. What is the greatest root of the equation $P(x) = 0$?



a. 4

b. 3

c. 1

d. 6

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24 [Foundational] • Target: Factoring higher-degree polynomials

The polynomial function $P(x) = 2x^3 + ax^2 - 5x + b$ has $(x - 1)$ as a factor. When $P(x)$ is divided by $(x + 2)$, the remainder is -18. What is the remainder when $P(x)$ is divided by $(x - 3)$?

a. 2

b. 8

c. 18

d. -30

25 [Foundational] • Target: Exponential and logarithmic equations

In the standard (x, y) coordinate plane, the graph of $y = f(x)$ contains the point $(-2, 5)$. The function g is defined by $g(x) = f(x - 3) - 4$. Which of the following points must lie on the graph of $y = g(x)$?

a. $(1, 1)$

b. $(-5, 1)$

c. $(1, 9)$

d. $(-5, 9)$

26 [Foundational] • Target: Function composition

Given the functions $f(x) = 3x - 4$ and $g(x) = x^2 + 5$, what is the value of $f(g(2))$?

a. 23

b. 9

c. 18

d. 27

27 [Foundational] • Target: Inverse functions

Solve $\log_2 x + \log_2 (x - 3) = 2$, and give the valid solution.

a. x=4

b. x=-1

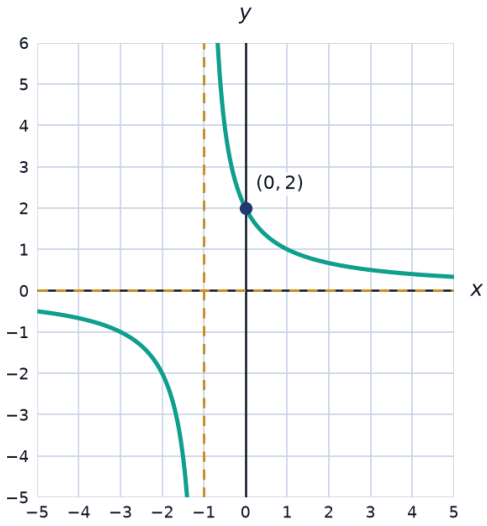
c. x=4,\ x=-1

d. x=7

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28 [Foundational] • Target: Rational equations

Let $P(x)$ be a polynomial of minimum degree with real coefficients such that $\lim_{x\to 1} \frac{P(x)}{x^2-3x+2} = 5$ and $\lim_{x\to 2} \frac{P(x)}{x^2-3x+2} = -3$. What is the value of $P(3)$?



a. -22

b. -11

c. 22

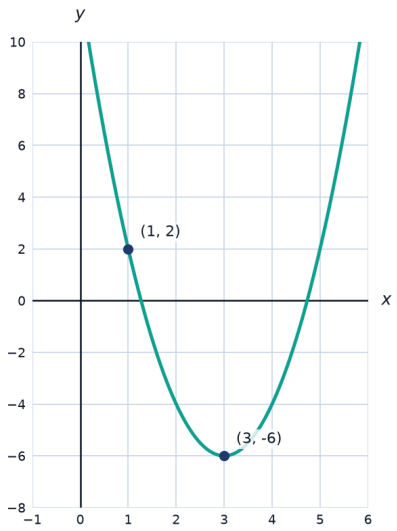
d. 42

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Factoring higher-degree polynomials**

The quadratic function $f(x) = ax^2 + bx + c$ is graphed in the standard (x, y) coordinate plane below. The vertex of the parabola is at $(3, -6)$, and the graph passes through the point $(1, 2)$. What is the value of b ?



a. -12

b. -6

c. 6

d. 12

30 **[Intermediate]** • **Target: Exponential and logarithmic equations**

The function f is defined by $f(x) = \frac{ax+8}{cx+d}$ for all real numbers $x \neq -\frac{d}{c}$, where a, c, and d are nonzero constants. In the standard (x, y) coordinate plane, the graph of $y = f(x)$ has a horizontal asymptote at $y = 2$, a vertical asymptote at $x = 3$, and an x-intercept at $(-4, 0)$. What is the value of $f^{-1}(4)$?

a. 10

b. 16

c. -2

d. 2

31 **[Intermediate]** • **Target: Function composition**

A function f is defined for all real numbers x such that $x \neq 0$ and $x \neq 1$. If $f(x) + 2f\left(\frac{1}{1-x}\right) = 3x$ for all permissible values of x, what is the value of $f(3)$?

a. $\frac{20}{9}$

b. $\frac{7}{3}$

c. $\frac{16}{9}$

d. 3

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32 **[Intermediate]** • **Target: Inverse functions**

Consider the function $f(x) = \frac{x^3 - 8}{x^2 - 4}$. We are interested in the behavior of this function near its discontinuities. Let $x = a$ be a value where the function has a removable discontinuity, and let $x = b$ be a value where the function has a non-removable discontinuity (vertical asymptote). What is the value of the limit of $f(x)$ as x approaches a, and what is the value of $\lim_{x \to b^-} f(x)$?

a. Limit at a is 3, $\lim_{x \to b^-} f(x) = -\infty$

b. Limit at a is 3, $\lim_{x \to b^-} f(x) = \infty$

c. Limit at a is 1, $\lim_{x \to b^-} f(x) = -\infty$

d. Limit at a is 1, $\lim_{x \to b^-} f(x) = \infty$

33 **[Intermediate]** • **Target: Rational equations**

Consider the rational function $f(x) = \frac{ax^2 - 4x + 1}{x^2 - bx + 2}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 2$. If $f(0) = \frac{1}{2}$, find the value of a.

a. 1

b. 2

c. 3

d. 4

34 **[Intermediate]** • **Target: Factoring higher-degree polynomials**

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

a. a = 5, b = 3

b. a = -5, b = 3

c. a = 5, b = 2

d. a = -5, b = 2

35 **[Intermediate]** • **Target: Exponential and logarithmic equations**

The function f is defined for all real numbers $xeq -2$ by $f(x) = \frac{4x - 1}{x + 2}$ What is the value of $f^{-1}(3)$?

a. -5

b. 3

c. 5

d. 7

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36 [Intermediate] • Target: Function composition

Consider the rational function $f(x) = \frac{ax^2 + bx + c}{dx^2 + ex + f}$, where a, b, c, d, e, f are non-zero constants. Given that $\lim_{x \to \infty} f(x) = \frac{1}{2}$ and $\lim_{x \to -1} f(x) = \infty$, determine the value of $\frac{a}{d}$.

a. $\frac{1}{2}$

b. 1

c. 2

d. Cannot be determined

37 [Intermediate] • Target: Inverse functions

The graph of the function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of the parameter a.

a. 1

b. 2

c. 3

d. 4

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Rational equations • Cross-Lesson: 1.01 Complex Numbers

The function f is defined by $f(x) = -x^2 + bx + c$, where b and c are constants. In the standard (x, y) coordinate plane, the maximum value of $f(x)$ is 14, and $f(0) = 5$. If $b > 0$, what is the value of $f(5)$?

a. 5

b. 10

c. 13

d. 14

39 [Challenging] • Target: Factoring higher-degree polynomials • Cross-Lesson: 2.01 Matrices & Vectors

Consider the rational function $f(x) = \frac{x^2 - 9}{x - 3}$. What is the value of $f(x)$ as x approaches 3?

a. 0

b. 6

c. Undefined

d. 3

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40 [Challenging] • Target: Exponential and logarithmic equations • Cross-Lesson: 3.01 Sequences & Series

Consider the exponential growth model described by the function $P(t) = P_0e^{kt}$, where $P(t)$ is the population at time t, P_0 is the initial population, and k is the growth constant. If the initial population P_0 is 500 and the population after 4 years is 2000, determine the value of the growth constant k to three decimal places.

a. 0.347

b. 0.416

c. 0.301

d. 0.477

41 [Challenging] • Target: Function composition • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Consider the rational function $f(x) = \frac{x^2 - 4}{x - 2}$. What is the value of the function as x approaches 2?

a. 4

b. 0

c. Undefined

d. 2

42 [Challenging] • Target: Inverse functions • Cross-Lesson: 6.01 Limits

Consider the exponential function $f(x) = 2^x$. If $f(a) = 8$, what is the value of a?

a. 1

b. 2

c. 3

d. 4

43 [Challenging] • Target: Rational equations • Cross-Lesson: 1.01 Complex Numbers

Consider the exponential function $f(x) = 2^x$. What is the value of $f(3)$?

a. 6

b. 8

c. 9

d. 5

44 [Challenging] • Target: Factoring higher-degree polynomials • Cross-Lesson: 2.01 Matrices & Vectors

Consider the rational function $f(x) = \frac{x^2 - 9}{x - 3}$. What is the value of $f(4)$?

a. 7

b. 1

c. 5

d. 13

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45 [Challenging] • Target: Exponential and logarithmic equations • Cross-Lesson: 3.01 Sequences & Series

A 13 m ladder leans against a vertical wall and reaches 12 m up the wall. Find the angle the ladder makes with the ground, to the nearest tenth of a degree.

a. 42.7°

b. 22.6°

c. 72.4°

d. 67.4°

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